



CLO 1.3

Construct and represent data using histograms, frequency polygons, bar graphs and pie graphs

BUS 1023: Applied Mathematics for Business

Monday, September 21, 2026



1. Graphs for Categorical Data – Bar Graph



A **bar graph** represents the data by using vertical or horizontal bars whose heights or lengths represent the frequencies of the data.

Example 1: College Spending for First-Year Students

The table shows the average money spent by first year college students. Draw a horizontal and vertical bar graph for the data.

Electronics	\$728
Dorm decor	344
Clothing	141
Shoes	72

Source: The National Retail Federation.



1. Graphs for Categorical Data – Bar Graph

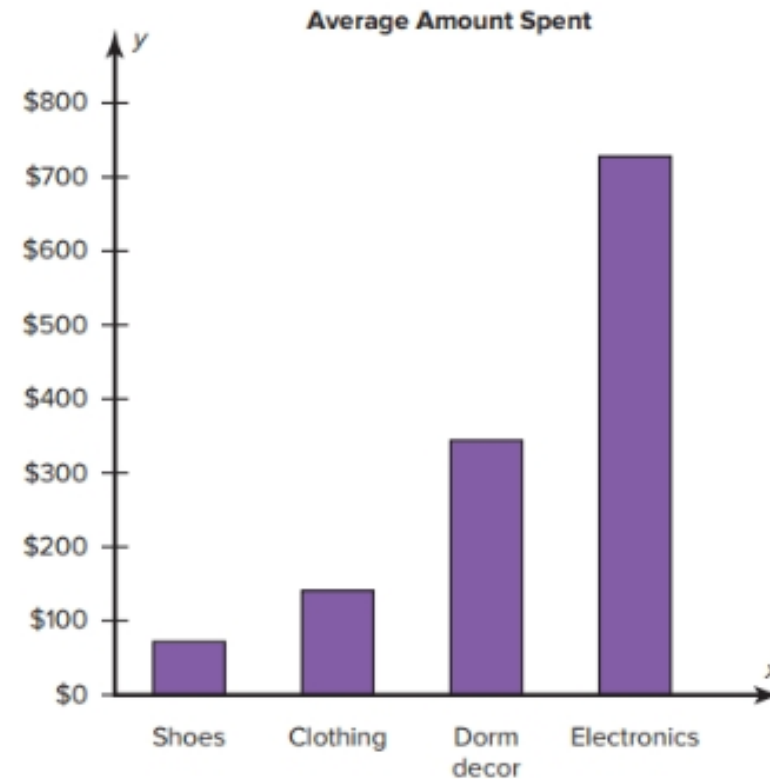
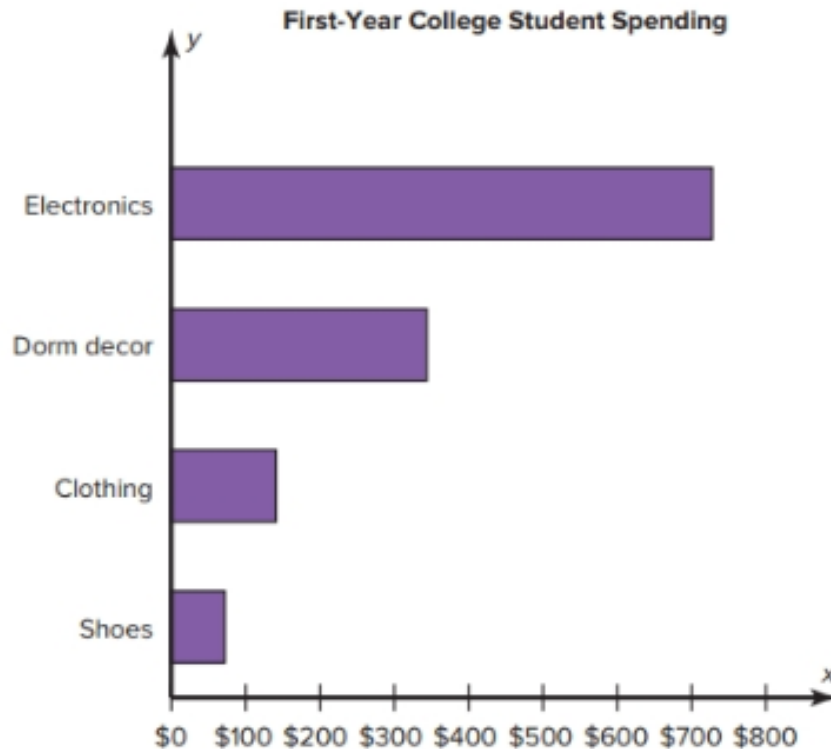
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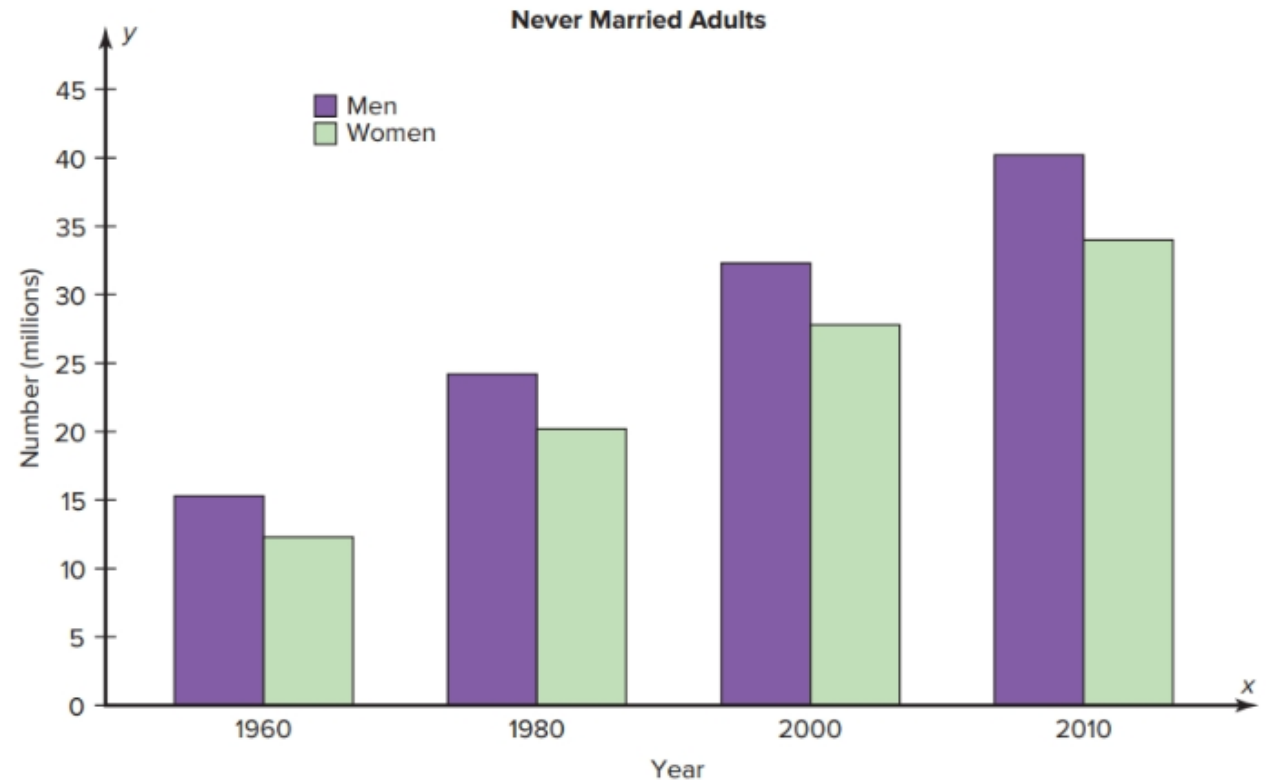


Bar graphs can also be used to compare data for two or more groups. These types of bar graphs are called **compound bar graphs**.

Example 2: Consider the following data for the number (in millions) of never married adults in the United States:

Year	Males	Females
1960	15.3	12.3
1980	24.2	20.2
2000	32.3	27.8
2010	40.2	34.0

Source: U.S. Census Bureau.

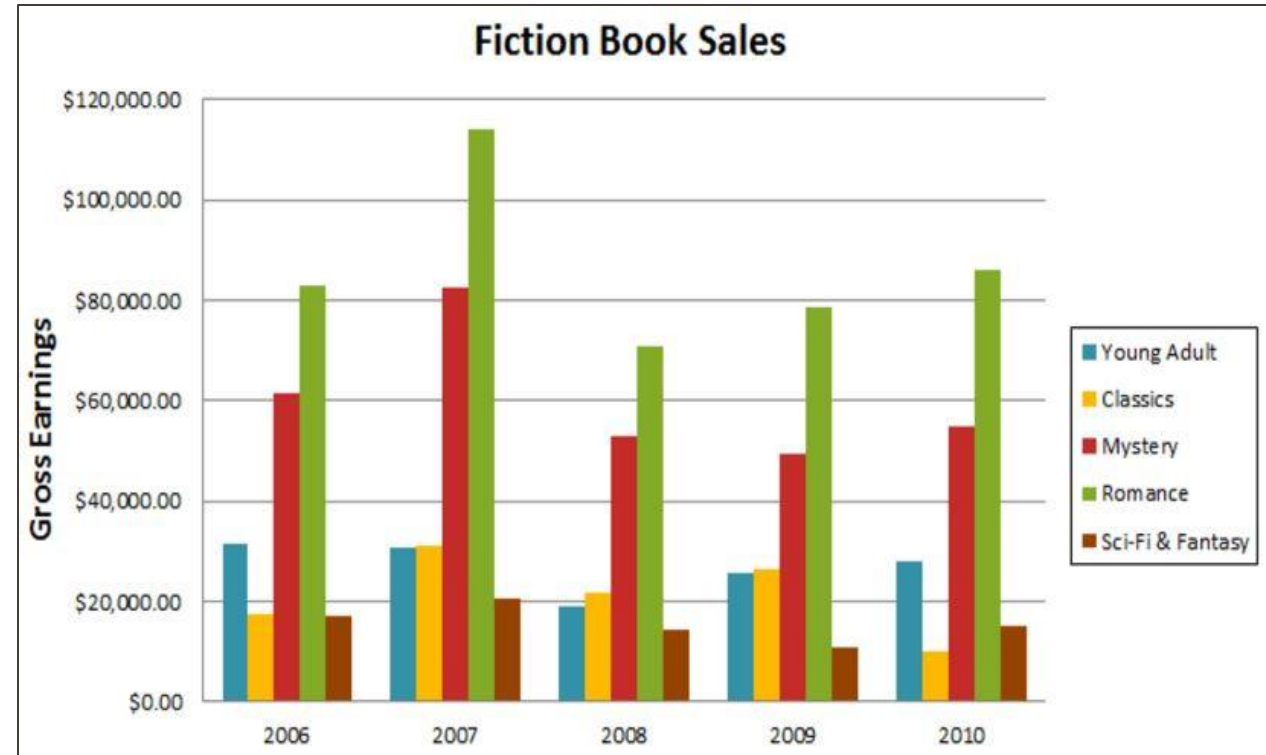




Interpretation of Compound Bar Graph:

Example 3: Complete the statements below by analyzing the data given in the graph.

- a) Romance was the most selling books every year.
- b) The "Young Adult" and "Classics" had very similar sales in year 2007 and 2009.
- c) The least selling book was 2010, in year Classics.
- d) "Mystery" had its highest sales in year 2007.
- e) In 2009, Young Adult and Sci-Fi and Fantasy had their lowest book sales.





2. Graphs for Categorical Data – Pie Graph

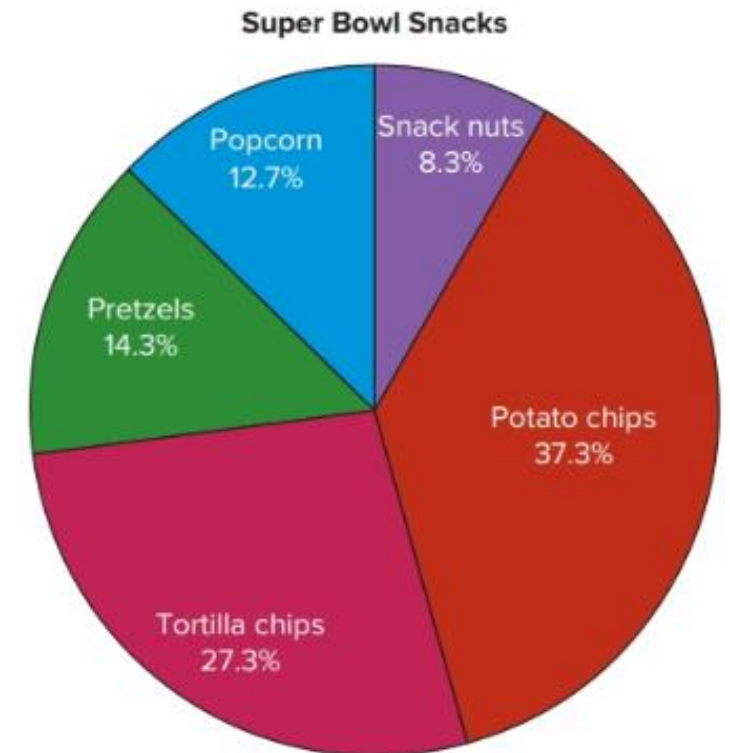


A **pie graph** is a circle that is divided into sections or wedges according to the percentage of frequencies in each category or the distribution.

Example 1: Super Bowl Snack Foods

This frequency distribution shows the number of pounds of each snack food eaten during the Super Bowl. Construct a pie graph for the data.

Snack	Pounds (frequency)
Potato chips	11.2 million
Tortilla chips	8.2 million
Pretzels	4.3 million
Popcorn	3.8 million
Snack nuts	2.5 million
Total $n = 30.0$ million	



Source: USA TODAY Weekend.



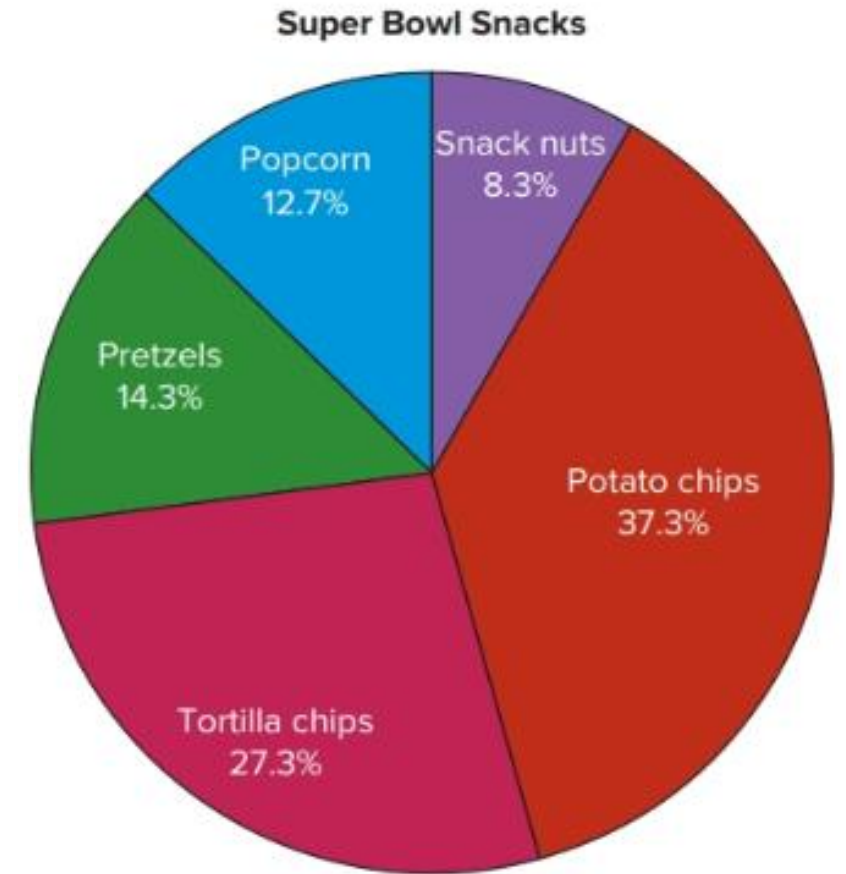
2. Graphs for Categorical Data – Pie Graph (Interpretation)

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Example 4: Complete the statements below by analyzing the data given in the graph.

- a) Which is the most favorite super bowl snack? **Potato chips**
- b) How many more pounds of potato chips were eaten compared to Popcorn?
7.4 million
- c) How many percent of the super bowl snacks were popcorn and snack nuts? **21 %**



N = 30 million



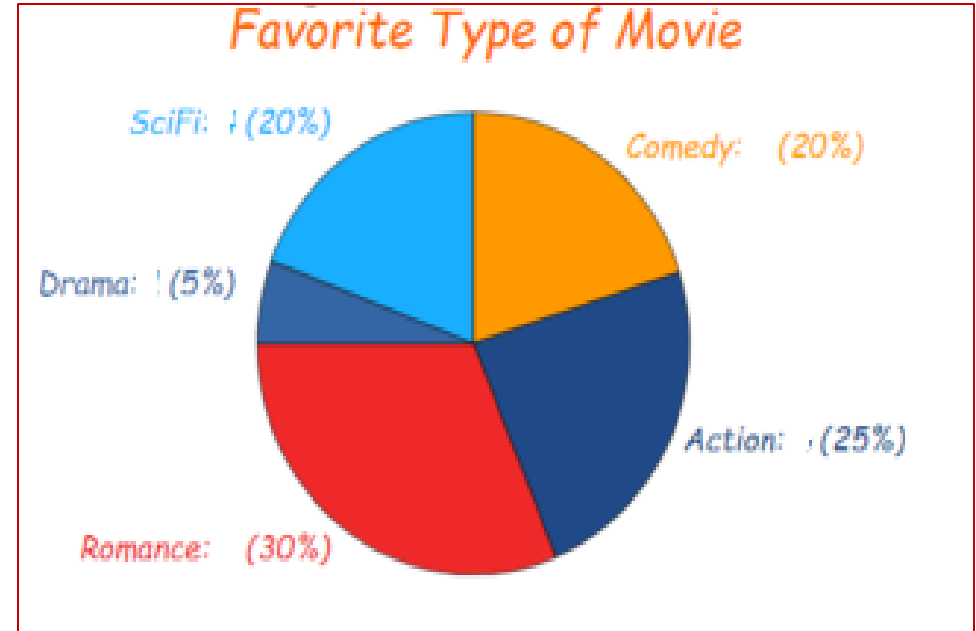
Example 5: A survey conducted on 200 people for their favorite type of movie is shown by the following Pie chart:

a) Which is the most favorite type of movie? **Romance**

b) How many people prefer the SciFi or action type of movie? $(20 + 25)\% \times 200 = 90$

c) How many more people prefer to watch comedy type of movie than drama type of movie? $(20 - 5)\% \times 200 = 30$

d) In another survey, there are 150 people who prefer Romance movies, how many people participated in that survey? $\frac{150}{30\%} = 500$



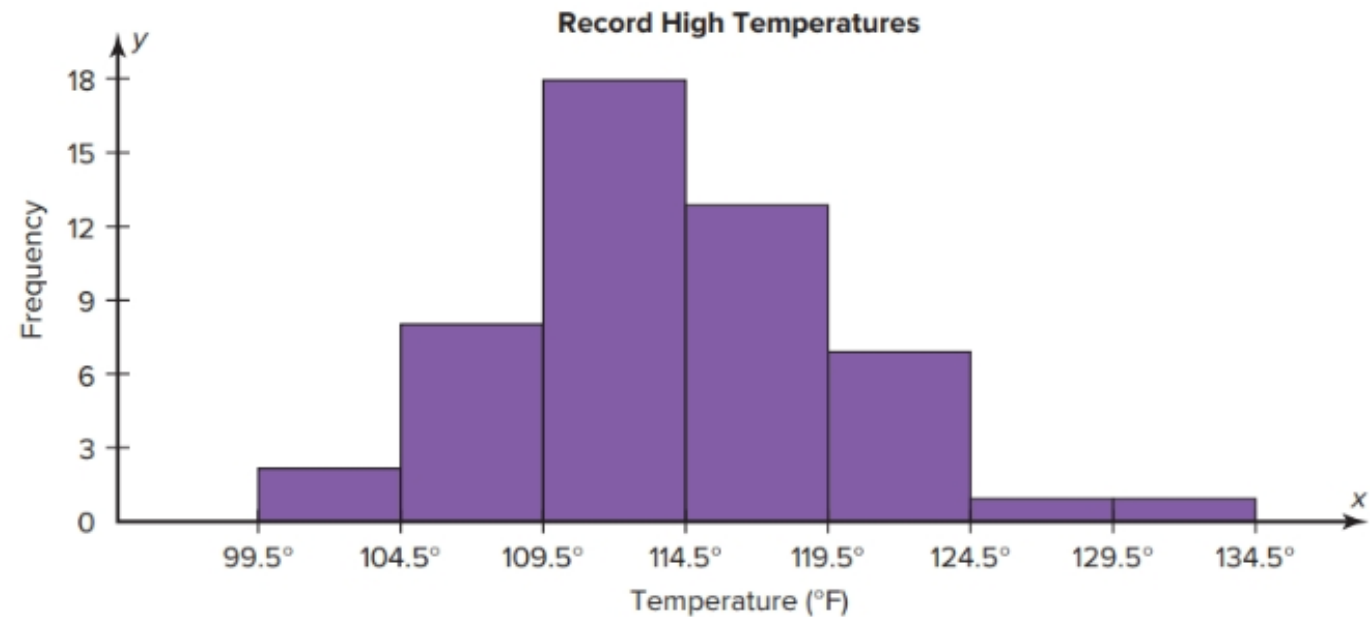
3. Graphs for Quantitative Data – Histograms

Histogram is a graph that displays the data by using vertical bars of various heights and widths to represent the frequencies of the classes. The **class boundaries** are represented on the horizontal axis.

Example 1: Record High Temperatures

Construct a histogram to represent the data shown for the record high temperatures for each of the 50 states

Class Limits	Class Boundaries	Frequency
100 - 104	99.5 - 104.5	2
105 - 109	104.5 - 109.5	8
110 - 114	109.5 - 114.5	18
115 - 119	114.5 - 119.5	13
120 - 124	119.5 - 124.5	7
125 - 129	124.5 - 129.5	1
130 - 134	129.5 - 134.5	1



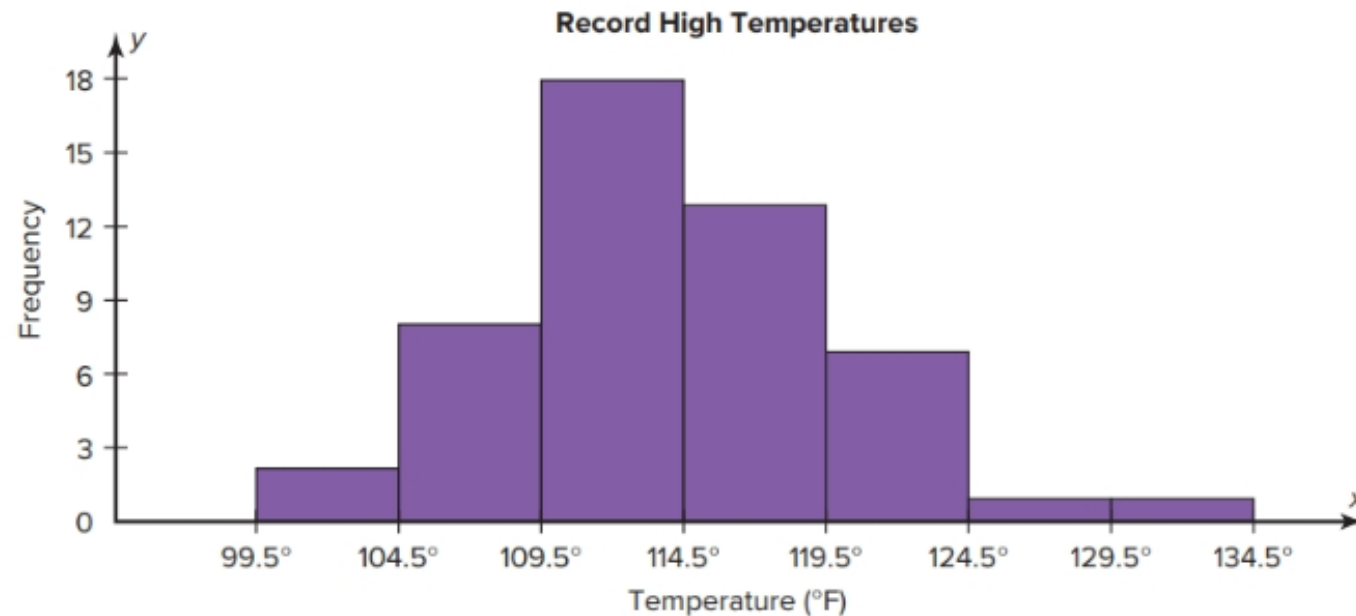


3. Graphs for Quantitative Data – Histograms (Interpretation)

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Example 6: The histogram below represents the temperatures for 50 states. Refer to the graph to answer the following questions:



1. How many states had the temperatures in the range 114.5 to 119.5°F?
Approximately 13 states
2. Which interval had the highest frequency? 109.5 – 114.5
3. How many states had temperature of at least 109.5 °F? _____
Approximately 40 states



4. Graphs for Quantitative Data – Frequency Polygon

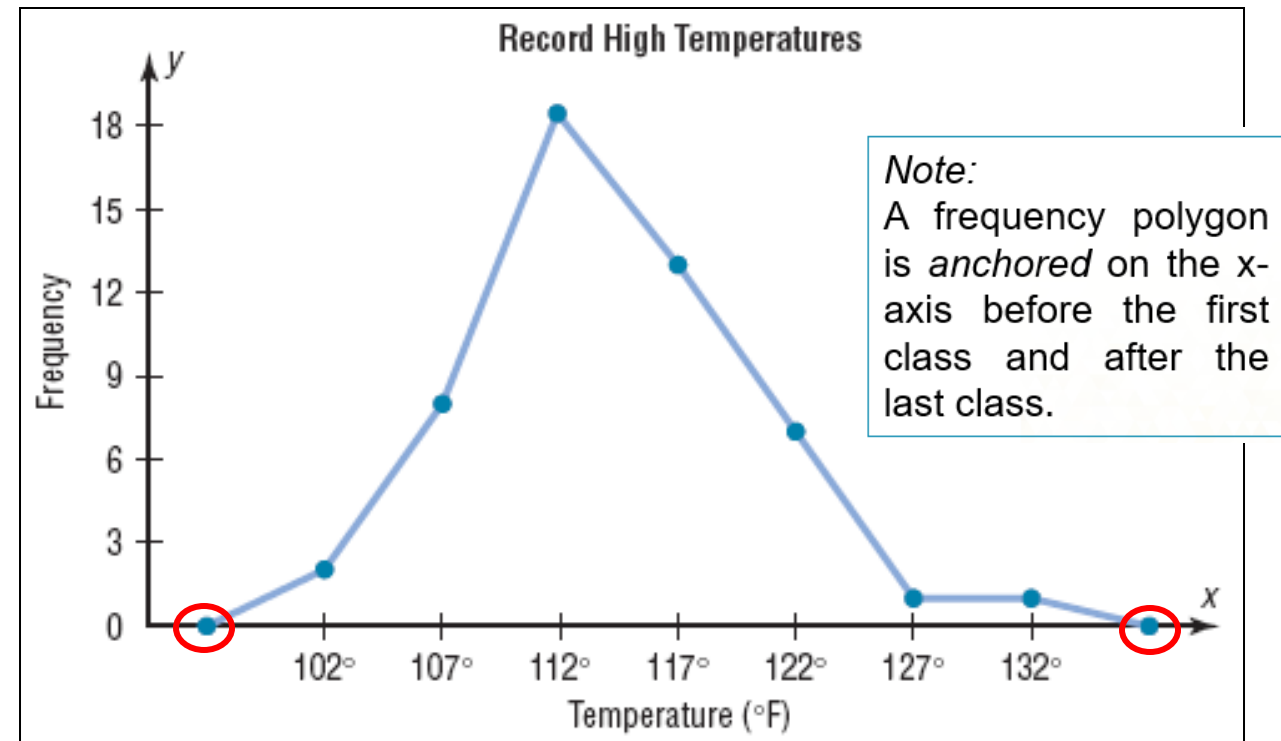
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Frequency polygon is a graph that displays the data by using lines that connect points plotted for the frequencies **at the class midpoints**. The frequencies are represented by the heights of the points.

To find the class midpoints average upper and lower class limit (add them and divide the result by 2). Class midpoints are represented on the horizontal axis.

Example 7: Construct a frequency polygon to represent the data for the record high temperatures for each of the 50 states

Class boundaries	Midpoints	Frequency
99.5–104.5	102	2
104.5–109.5	107	8
109.5–114.5	112	18
114.5–119.5	117	13
119.5–124.5	122	7
124.5–129.5	127	1
129.5–134.5	132	1





4. Graphs for Quantitative Data – Frequency Polygon (Interpretation) 12

Example 1: The graph below represents the data for the record high temperatures for each of the 50 states

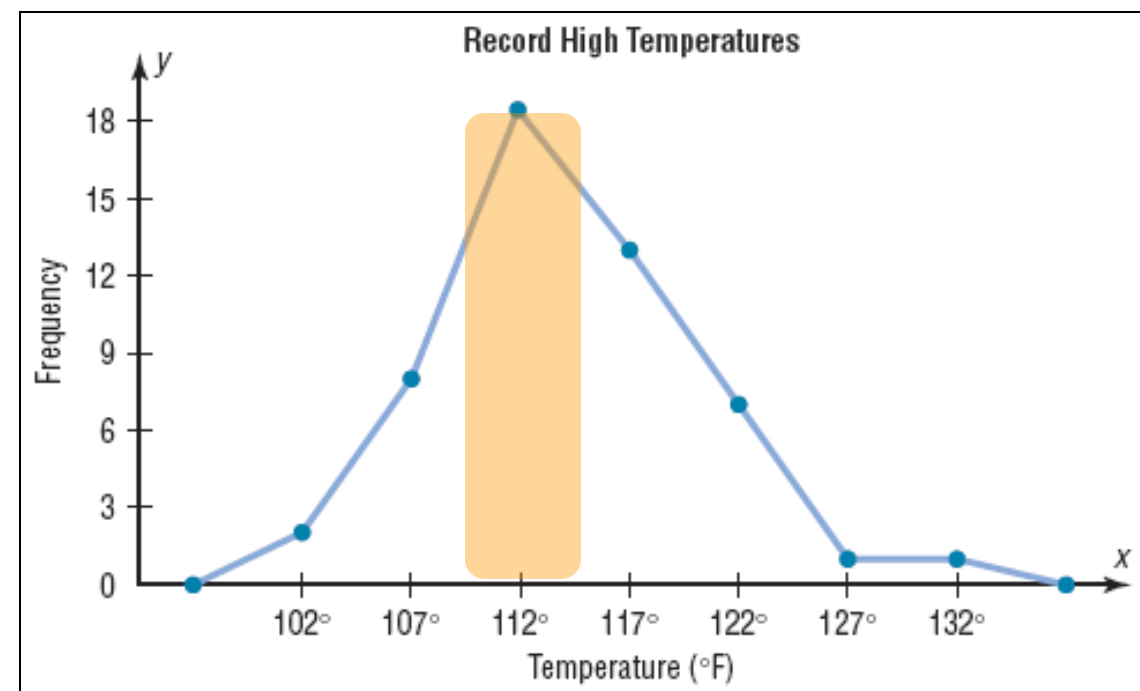
a) What was the highest temperature recorded? 132 °F

b) What temperature was the most common among the 50 states? 112 °F

c) How many states have temperature of at least 122 °F degrees?

Approximately 9 (7+1+1)

d) What percentage of states have at most 112 degrees? $\frac{28}{50} \times 100 = 56\% \text{ or approx. } 56\%$



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